Jackie Dai

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Education

University of California, Berkeley — Berkeley, CA

Bachelor of Arts, Double Major in Computer Science and Statistics — *Expected December 2026*

- **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Database Systems, Computer Graphics, Computer Vision, Internet Architecture, Software Engineering, Game Design & Development, Computer Security, Programming Languages & Compilers

Experience

Software Engineer Intern — Tokyo, Japan

YuaBread, Inc. — *May 2025 – August 2025*

- Built and deployed a full-stack manufacturing-planning platform that processed orders from PDFs, email, and Slack, reducing spreadsheet processing time from **6 hours to 30 minutes** and eliminating manual-entry errors.
- Implemented the platform using **Next.js, FastAPI, and PostgreSQL** and implemented monitoring, authentication, and role-based access control

Software Engineer Intern — Singapore, Singapore

Broctagon — *May 2024 – August 2024*

- Built a AI-powered forex trading advisor using **AWS Bedrock, AWS Lambda, Python, and MongoDB**, enabling the agent to retrieve market data through tools, supporting users with more informed trading decisions.
- Designed the interface in **Figma** and implemented it in **React.js**, integrating the backend agent to support user queries and display model output.

Undergraduate Research Assistant — Berkeley, CA

UC Berkeley EECS — *January 2025 – May 2025*

- Developed a mixed reality stroke rehabilitation platform using **Unity, C#, and Meta Quest 3**, enabling therapists to place and edit virtual objects for rehabilitation exercises.
- Integrated cross-platform video calling between therapist laptops and patient VR headsets.
- Collaborated with **Stanford and UCSF** to evaluate the prototype with stroke survivors and co-authored a research paper on mixed reality stroke rehabilitation.

Course Instructor — Berkeley, CA

Game Design and Development at UC Berkeley — *May 2023 – May 2026*

- Taught weekly lectures on **Unity, C#, Git,** and game design to classes of $\tilde{4}0$ undergraduate students.
- Led a team of 3 in designing and developing new programming projects and course curriculum for students.
- Designed, developed, and maintained the course website to support instructional materials and student resources.

Projects

Secure File Sharing System | *Go, Cryptography, Git* — February 2024 – March 2024

- Built an encrypted file-sharing system in **Go** with password-based authentication, secure file storage, and integrity verification using Argon2 key derivation and message authentication codes.
- Implemented invitation-based sharing using public-key encryption and digital signatures, with cascading access revocation to remove users and their downstream recipients.
- Designed chunk-based storage to append data without downloading existing file contents; wrote tests for tampering, unauthorized access, sharing, and bandwidth efficiency.

Rate My Classes | *Next.js, React, Python, NextAuth.js, Git* — September 2023 – December 2023

- Built a full-stack course review platform allowing students to rate and review UC Berkeley classes.
- Implemented authentication, protected routes, user accounts, and responsive frontend interfaces.
- Designed backend APIs and database models for retrieving course data scraped from the web.

Technical Skills

Languages: Python, **Rust**, C, C++, C#, Go, Java, JavaScript, TypeScript
Frontend: React, Next.js, Vue.js, Three.js, HTML/CSS
Backend: SQL, NestJS, FastAPI, REST APIs, PostgreSQL, MongoDB
Developer Tools: Git, Docker, Make, AWS, Postman, Google Cloud, Unity, Unreal Engine, Figma